5.3.1 The general equation of a Conic

Let $S(x_1, y_1)$ be the focus, l the directrix, and e be the eccentricity. Let $P(x, y)$ be the moving point.

By the definition of conic, we have

$$\frac{SP}{PM} = \text{constant} = e, \qquad \ldots(1)$$

Where $SP = \sqrt{(x-x_1)^2 + (y-y_1)^2}$

and PM = perpendicular distance from $P(x, y)$ to the line $lx + my + n = 0$

$$= \left| \frac{lx + my + n}{\sqrt{l^2 + m^2}} \right|.$$

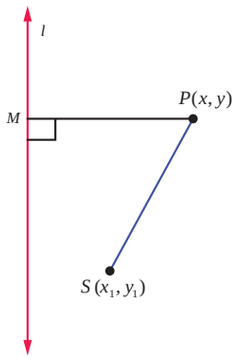


Fig.5.17

From (1) we get $SP^2 = e^2 PM^2$

$$\Rightarrow \quad (x-x_1)^2 + (y-y_1)^2 = e^2 \left[\frac{lx + my + n}{\sqrt{l^2 + m^2}} \right]^2.$$

On simplification the above equation takes the form of general second-degree equation $Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$, where

$$A = 1 - \frac{e^2 l^2}{l^2 + m^2}, B = \frac{2lme^2}{l^2 + m^2}, C = 1 - \frac{e^2 m^2}{l^2 + m^2}$$

Now,

$$B^2 - 4AC = \frac{4l^2 m^2 e^4}{\left(l^2 + m^2\right)^2} - 4\left(1 - \frac{e^2 l^2}{l^2 + m^2}\right)\left(1 - \frac{e^2 m^2}{l^2 + m^2}\right)$$

$$= 4\left(e^2 - 1\right)$$

yielding the following cases:

(i) $B^2 - 4AC = 0 \Leftrightarrow e = 1 \Leftrightarrow$ the conic is a parabola,

(ii) $B^2 - 4AC < 0 \Leftrightarrow 0 < e < 1 \Leftrightarrow$ the conic is an ellipse,

(iii) $B^2 - 4AC > 0 \Leftrightarrow e > 1 \Leftrightarrow$ the conic is a hyperbola.

5.3.2 Parabola

Since $e = 1$, for a parabola, we note that the parabola is the locus of points in a plane that are equidistant from both the directrix and the focus.

(i) Equation of a parabola in standard form with vertex at (0, 0)

Let S be the focus and l be the directrix.

Draw SZ perpendicular to the line l.

Let us assume SZ produced as x-axis and the perpendicular bisector of SZ produced as y- axis. The intersection of this perpendicular bisector with SZ be the origin O.

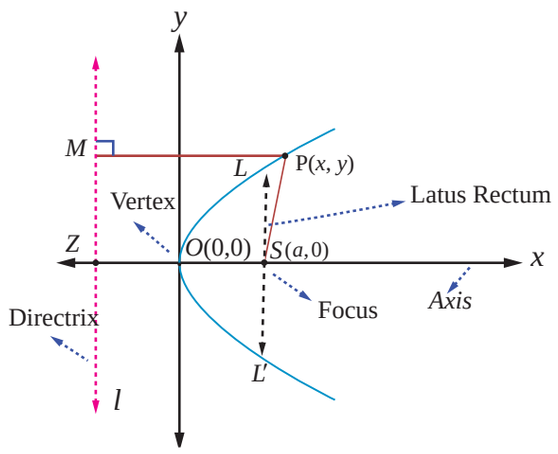


Fig.5.18

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